

Sanya Gupta

+91 8288 971449 | guptasanya366@gmail.com | [LinkedIn](#) | [GitHub](#) | [LeetCode](#)

EDUCATION

University of Engineering & Technology, Panjab University

Bachelors of Engineering in Information Technology | CGPA 8.18 SGPA 9.11

Chandigarh

July 2023- August 2027

GGMSSS 20B

PCM CBSE BOARD | 93%

Chandigarh, India

April 2021- April 2023

TECHNICAL SKILLS

Generative AI & LLMs: LLMs (Transformers, Hugging Face), RAG (LangChain, FAISS), Computer Vision (OpenCV,YOLO), TensorFlow, PyTorch, scikit-learn, NumPy, Deep Learning (VNET, UNET, GAN, Autoencoder)

Backend & APIs: Python, FastAPI, Flask, RESTful APIs, JWT/OAuth 2.0, Asyncio

Database: PostgreSQL (JSONB/Vector storage), MySQL, pgvector

Frontend & Design: JavaScript, React, Tailwind CSS, Data Visualization.

Tools & DevOps: Docker, CI/CD, Linux, Git, Google Cloud Platform (GCP), Jupyter, MATLAB, C++.

EXPERIENCE

AISoC [AI Summer of Code]

June 2025 - July 2025

Open Source Contributor | Python, Deep Learning, YOLO

- Deployed a real-time vehicle detection system using **YOLO**
- Optimized YOLO-based inference for low-light environments, improving precision to 0.82 through hyperparameter tuning

Design Innovation Center, PU

May. 2024 – Jan. 2025

Deep Learning Intern | Python, DL architectures

- Spearheaded experimentation with **3D Architectures (U-Net, V-Net, and GANs)** on a novel skull dataset, optimizing the training loop to significantly reduce computation time
- Developed an automated Python pipeline leveraging Connected Component Analysis (CCA) to refine segmentation masks, entirely eliminating noise and significantly boosting model precision.

PUBLICATION

[SCAI-NET](#) - An AI-Driven Framework for Optimized, Fast, and Resource-Efficient Skull Implant Generation for Cranioplasty using CT Images, published in Journal of Computers in Biology and Medicine [CS-13, Impact Factor: 7]

- Developed **SCAI-Net**, an AI-driven framework for fully automated cranial implant design from CT scans with minimal post-processing.
- Built and validated a dual-pathway system generating accurate, ready-to-3D-print implants in ~93.5 seconds.

PROJECTS

Cite Craft | [website](#) | React, FastAPI, PostgreSQL, RAG, Transformers

- Designed a section-aware RAG tool to improve researcher productivity. Prioritized feature development for 'Source Attribution' based on user need for 100% verifiability in AI-generated reports
- Developed a multi-source aggregator** integrating **5 academic APIs** (including Semantic Scholar and arXiv); engineered a data pipeline for automated metadata extraction and **CSV-based data portability** for large-scale literature reviews.
- Engineered a **comparative analysis engine** featuring per-cell RAG retrieval; implemented **direct source attribution** to display original paper snippets, ensuring 100% verifiability of AI-generated comparisons.
- Orchestrated a high-performance backend with **FastAPI** and **PostgreSQL (JSONB)** for efficient embedding storage; implemented **JWT-based secure access** and managed a multi-cloud deployment via Railway and Vercel.
- Optimized for high-performance retrieval, achieving **sub-500ms response times** for RAG-based queries through efficient **PostgreSQL JSONB indexing** and asynchronous FastAPI endpoints.

AutoOutreach | [code](#) | Python, Google OAuth, Hugging Face

- Developed an **automated lead generation and outreach engine** using Python, integrating **Google OAuth 2.0** and **Google Sheets API** for dynamic data synchronization.
- Leveraged **Hugging Face Inference APIs** to perform zero-shot analysis of job descriptions, generating highly personalized email bodies via tailored prompt templates.